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International Mobility and Social Capital in the Academic Field

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Abstract The relationship between the international mobility of academic researchers and social capital is complex. On the one hand, the literature suggests that social capital facilitates the international mobility of academics which, in turn, promotes the accumulation of international social capital, enhances research productivity, and advances careers. On the other hand, international mobility can isolate researchers from the national social capital in their origin countries. In this paper, I present the results of 42 interviews in Canada and Germany to examine how academics in both countries have experienced the connection between international mobility and social capital. In addition to revealing the complexity of this connection, the results show that social capital facilitates international mobility and that mobility sometimes creates social capital. However, mobility can also lead to the loss of national social capital that negatively affects early-career researchers in particular.

Keywords Academic researchers · International mobility · Social capital · Canada · Germany

Introduction

Academic research thrives on personal interaction, communication, and cooperation. In this context, international mobility can be crucial for productivity and for accumulating career-enhancing social capital. This paper contributes to an ongoing debate related to international academic mobility (e.g., Jöns 2006, 2008; Kim

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2009; Uusimaki and Garvis 2017), specifically concerning the connection between international mobility and social capital. In the academic field, it is unclear whether international mobility has positive or negative effects related to social capital. On the one hand, international mobility can enhance international connections and provide access to resources, including jobs and prestige (Burris 2004). On the other hand, mobility can be associated with the loss of domestic connections in the country of origin and thus a reduction of national social capital (David et al. 2010). This conundrum raises important questions for the development of policies and institutional strategies related to the international mobility of academics: Does existing social capital lead to greater international mobility or does international mobility create social capital? Does international mobility enhance careers or does it impede career development when national social capital in the country of origin or a previous stay is lost? To address these questions, I conducted an empirical study involving academic researchers in Canada and Germany, exploring the ambiguous relationship between international mobility and social capital. In this paper, I present the results of this study.

For the purpose of this paper, I define "social capital" as networks and ties that provide access to particular social or professional communities (Granovetter 1973; Portes 1998). In this context, Mark Granovetter (1973) distinguishes between weak and strong ties, and suggests that casual weak ties offer better access to professional opportunities than more intimate strong ties. I chose this definition because I expect networks and ties to play similar roles in the academic field. The literature on higher education and the scientific process often frames social capital in terms of social and professional networks that structure research collaborations (Bozeman and Corley 2004), and social ties that extend to other universities, research groups, or labs (Pezzoni et al. 2012). Social capital also structures the "invisible college" consisting of tightly-networked research groups and communities that collaborate and that can control access to resources and information (De Solla Price and Beaver 1966; Wagner 2009). A Bourdieuian perspective of social capital adds that membership in a group provides access to resources that facilitate the professional reproduction of this group. If this community is confined to a particular country, then I speak of national social capital. Conversely, international social capital transcends national borders. Like Bourdieu (1988), I focus on the academic field (Bourdieu called it the "university field"), referring to a distinct social setting in which particular social capital has value. In this field, I examine how academic researchers experience the connection between international mobility and social capital. "Academic researchers" include, in the context of this paper, researchers usually based at institutions of higher education, ranging from doctoral students and postdoctoral fellows to full professors. Finally, in the literature on mobile academic researchers, the term "international mobility" tends to encompass a range of different types and activities, including obtaining a degree abroad, short-term work-related visits, longer-term professional stays, permanently moving to a foreign country for an academic job, and developing an academic career after entering a country as student (Rostan and Ceravolo 2015); it involves circular and one-way movements (Jöns 2009); and it can be measured using different indicators, such as citizenship, country in which a degree was earned, time spent abroad, and career stage when mobility occurs

(Teichler 2015). For the purpose of this paper, I define international mobility in a more limited way that may come close to what some research describes as "migration" (Teichler 2015: 11): as cross-border moves that are either permanent or extend over a period beyond one year. By excluding foreign short-term research and sabbatical visits and by focusing on long-term stays, I expect that international mobile academics have opportunities to accumulate social capital in the form of both weak ties, which can be acquired through casual encounters, and strong ties, which may require more sustained exposure to a community.

In the following sections, I first review existing research on the connections between international mobility and social capital. Then, I explain the methods of the empirical investigation; thereafter, I present the results. In the Conclusion, I interpret the results in light of the literature and their policy implications, and make suggestions for further research.

Social Capital and International Mobility in the Literature

Social capital can generally facilitate labor market entry and success (Granovetter 1973). It can also be an important resource available to international migrants and their successful settlement and economic integration in destination countries (Bauder 2006; Valade and Tyyskä 2019). In the context of the international mobility of academic researchers, the literature offers variable interpretations of the effects of social capital. These interpretations can broadly be categorized as "optimistic" and "pessimistic." The former proposes that the connections between international mobility, social capital, and subsequently professional development are associated with positive experiences; the latter presents these connections in a negative light.

The optimistic interpretation involves at least three narratives: the first narrative suggests that *social capital facilitates international mobility* of academic researchers. Social capital in the form of "existing personal networks of supervisors or mentors" (Schiller and Diez 2012: 1330) guides researchers' towards particular locations abroad. Scientific foundations, such as Germany's Alexander von Humboldt Foundation or the USA's Fulbright Foundation, and the networks they cultivate facilitate the international mobility and exchange of researchers often drawing on existing networks (Chepurenko 2015; Jöns 2009). There are differences across scientific fields in the way in which social capital facilitates mobility: in Nordic countries, international visits among doctoral students who work towards degrees in natural science disciplines (e.g., Physics, Chemistry) are often arranged through the supervisor and follow the supervisor's network; in the social sciences, international mobility tends to be initiated by the students themselves (Kyvik et al. 1999). Institutional contexts may also be important in the way social capital translates into international mobility. For example, a supportive environment promoted by chairs, senior colleagues, and peers can encourage faculty members to use their networks and engage in short-term international mobility or participate in international faculty exchanges (Patrício et al. 2018). These environments are shaped by an institution's "competitive horizon" (Hoffman et al. 2008) that can portray internationalization either as an opportunity or a constraint. In the context of long-term international mobility, which is

often associated with a change of employer, international mobility paths may be more influenced by personal aspirations and career-building strategies (Jung 2014; Khattab and Fenton 2016) and facilitated by personal networks and career aspirations (Bennion and Locke 2010). Nevertheless, institutional environments are still important to the way in which social capital translates into mobility. For example, faculty hires often occur within networks and hierarchies of prestigious universities and department (Amir and Knauff 2008; Clauset et al. 2015). These networks often extend beyond national boundaries (Amir and Knauff 2008).

Another narrative related to the optimistic interpretation suggests that *international mobility develops social capital* by fostering new connections and/or enhancing existing connections among academic researchers. Decades ago, Derek De Solla Price and Donald Beaver (1966: 1017) observed, in the context of the invisible college, an acceleration of research collaboration and co-authorships "deriving from the increased mobility of scientists." Through international mobility, scientists continue to obtain an "opportunity to gain access to international knowledge networks" (Schiller and Diez 2012: 1330). The experiences academic researchers acquire abroad can establish international connections that provide access to resources for successful knowledge production and innovation (Casanueva and Gallego 2010; Patrício et al. 2018), increase research productivity, and lead to outputs, such as high-impact publications with international co-authors (Jonkers and Cruz-Castro 2013). In this way, the social capital that academic researchers develop through international mobility can contribute to the accumulation of other forms of capital, such as enhanced skills (human capital), publications in prestigious journals (objectified cultural capital), or well-regarded international credentials (institutionalized cultural capital). The evidence related to the issue of whether international mobility generates productive social capital, however, is ambiguous. For example, "weak ties," i.e., casual professional acquaintances (Granovetter 1973), tend to be more beneficial to researchers' scientific performance than "strong ties" (Contandriopoulos et al. 2016), which raises the question if academics without long-term stays abroad could also develop these ties through attending international conferences, exchanges with international visitors, or online communication. For example, returnees who studied overseas and now work in the higher-education systems of Korea, Hong Kong, and Malaysia tend to not have higher research productivity than their colleagues who studied domestically (Shin et al. 2014).

Following from this second narrative, the third narrative to be examined is whether *social capital accumulated abroad facilitates career advancement* through the acquisition of other forms of capital that increase an academics' market value. International social capital plays an "often hidden role" (Waters 2009: 116) in valorizing other forms of capital. In Hong Kong, networks and connections that mobile academics established to Vancouver, Canada, have amplified the value of credentials (i.e., institutionalized cultural capital) from the Vancouver-based University of British Columbia, propelling the careers of academics who possess such credentials. The impact of international social capital, however, depends on a range of factors. Research using Mexican data shows that the effect of social capital varies between disciplinary knowledge fields, a researchers' position in a network, and the extent of the network (Gonzalez-Brambila 2014). Whether the social capital accumulated

abroad consists predominantly of weak or strong ties (Granovetter 1973) may also affect how this social capital translates into career advancement (Contandriopoulos et al. 2016). Furthermore, international mobility is more common among early-career and younger researchers than among established and older academics (Auriol 2010). International social capital that enhances the 'market' value of researchers tends to be accumulated in the early stages of academic careers (Scheibelhofer 2006). Nevertheless, if international social capital accumulated in early career stages can be maintained over time and over geographical distances, then it can exert lasting positive effects on academic careers. A study of mobile academics in Singapore, however, suggests that social capital accumulated abroad may be difficult to maintain over geographical distance after a move away (Wang et al. 2019).

The pessimistic interpretation relates to a literature that emphasizes the negative side of international mobility. One aspect of this negative side emphasizes the structural inequalities and the hierarchical and asymmetrical nature of the international academic system (Bauder et al. 2018). Prestigious institutions in the global north are particularly attractive to postdocs and other researchers from the global south, especially from emerging Asian knowledge economies (Franzoni et al. 2012; Kim 2009; Paul 2018; Khattab and Fenton 2016). Nevertheless, recent research also observes migration from the geographical core to the periphery of academic knowledge production (Lee and Kuzhabekova 2018). Some national university systems, such as Singapore's, assume intermediate positions (Ortiga et al. 2019). A hierarchy also exists among the countries of the global north (e.g., Morano-Foadi 2005). These structural inequalities and hierarchies have produced and reinforced the hegemonic positions of the leading centers of knowledge production in the global north (Jöns 2015) and supply contingent academic labor to these centers (Cantwell 2011a; Khattab and Fenton 2016). A related aspect focusses on negative personal effects of international mobility. For postdoctoral researchers, in particular, international mobility can be an undesired outcome of lacking domestic opportunities and coercion, and be associated with discrimination and exploitation (Ackers 2008; Cantwell 2011b). Göran Melin (2005) finds that some internationally mobile scholars who return to Sweden after a postdoctoral period abroad do not benefit from the foreign experience. Although "postdocs are carriers of new knowledge and agents in social networks" (Melin 2005: 236), the institutional cultures at their current workplace are not always receptive to this knowledge and these networks. In the context of this paper, the pessimistic interpretation of the connections between international mobility and social capital suggests that social capital is relevant predominantly within the national – rather than the international – academic field. In this case, the "competitive horizon" may be national rather than global in extent (Hoffman et al. 2008). According to this interpretation, having strong national academic ties is more important than possessing an extensive international network. International mobility may thus be a disadvantage for career advancement relative to staying put.

This pessimistic interpretation is based on the particularity of the *national academic field*. International mobility – and to which country and under what circumstances mobility occurs – can play different roles and is valorized differently in different national academic fields (Bauder et al. 2017). For example, in Japan, academics who earned their doctorate degree abroad serve "as gatekeepers" of

scientific communication because they converse "more frequently with their international peers" than their peers with Japanese doctorate degrees (Horta and Yonezawa 2013: 546). Conversely, in the USA, foreign faculty members are more productive in research relative to their non-migrant peers, suggesting that their role is increased knowledge production (Mamiseishvili and Rosser 2010). Furthermore, there are national differences in which career stage international mobility signifies value: in Australia's national higher-education system, there tends to be a greater preference for internationally mobile staff at higher ranks and ages compared to the USA where international mobility is more appreciated at a mid-career level (Welch 1997: 332).

Pierre Bourdieu's (1988) work on the French academic system further illustrates that academic hierarchies and the value of social and other forms of scientific and academic capital are national in extent. The importance of social capital within the national academic field is well-documented in the literature. In Germany, becoming a tenured professor in Sociology is not only a factor of scholarly output and merit but it is also a matter of the size of a person's network and reputation within the national academic field (Lutter and Schröder 2016). Heinke Röbken (2009: 233) suggests that in Germany "mobility and social networks can have a profound effect on the employment chances of young academics in the labor market," whereby mobility does not refer to international but *national* mobility. Research shows that foreign applicants to professorships in the Humanities in Germany are perceived to have diminished chances for being hired because they lack national social capital; their experiences suggests that "it is almost impossible for scholars without or only loose network connections to Germany to obtain a professorship at a German university" (Behrens et al. 2010: 68, my translation). In the Greek academic system, Nikos Bozionelos (2014: 288) observes that "social capital determines careers within that system. … [S]ocial capital or know-whom is not just a contributing factor …, but instead a sufficient condition for career entry and success within that labor market." In Singapore, foreign scientists strategically accumulate and deploy various forms of capital, including social capital, to secure positions (Sidhu et al. 2015). Another study illustrates how international academics locating in Singapore tend to quickly increase research collaborations with local colleagues while their international collaborations decline, suggesting that these academics focus on cultivating national social capital in light of the difficulty of maintaining international social capital over long distances (Wang et al. 2019).

There are important differences between countries regarding the way in which national social capital matters for career progression. While in France an affiliation with a public research organization is important for career progression, in Italy connections to senior members within a discipline or university tend to be more useful (Pezzoni et al. 2012). In Germany, "personal contacts and loyalty vis-à-vis decision makers" are sometimes perceived as more important for acquiring a professorship than the quality of the applicant (Behrens et al. 2010: 68, my translation). Other forms of capital that are also particular to a national academic field interlock with national social capital. Studying "identity capital" validating group membership among internationally-mobile academics, Terri Kim (2017: 993) observes that "the rules of stratification in academic capitalism are not transnational but 'place-bound'." Members of the Chinese scientific diaspora cite the absence of

cultural-particular social capital in the form of *Guangxi* as an advantage of working in Australia (Yang and Welch 2010).

That the academic field and the corresponding social capital is national in scale is in itself not negative. For example, national networks may help Chinese doctoral graduates to stay in Australia (Yang and Welch 2010). However, researchers may also experience a loss of connections and the diminishment of their networks in the national academic field of their origin when they move abroad. In this case, international mobility is associated with a *loss of national social capital*, which can have negative career impacts. This situation may occur especially in Germany. Daniel Schiller and Javier Revilla Diez (2012) find that German scholars who have been abroad for more than three years are cultivating international networks "at the expense of linkages within the German community … The group that stayed abroad for shorter periods is also oriented towards the international community, but … they have better contacts at the national level" (1327). Recent research corroborates that social capital accumulated in a particular country diminishes after an international move because "relationships require maintenance, which necessitates time and energy" (Wang et al. 2019: 458). Longer stays abroad especially reduce national social capital.

Methods

To empirically examine the various narratives of how international mobility and social capital are connected, I conduced interviews with internationally-mobile academics in Canada and Germany. This choice of countries reflects different expectations based on the literature with respect to the role of social capital related to the international mobility of academic researchers. Although the higher education systems in both countries are highly advanced and very attractive to internationally mobile researchers – especially in their early-career stage – from the global south, there are important differences between the two countries. The proportion of internationally mobile academics tends to be higher in Canada than in Germany across a range of indicators (Teichler 2015). Canada seeks to attract elite foreign academics, for example, through the Canada Research Chair program. Since the Canadian academic system is closely integrated with that of the USA, one can expect that the role of foreign faculty members is increased knowledge production (Mamiseishvili and Rosser 2010). Conversely, German organizations, such as the Alexander von Humboldt Foundation, fund not only incoming scholars but also encourage German academics, especially in their early-career stages, to spend time abroad, suggesting that there is a perception of benefits derived from two-way international exchanges. In fact, international researchers staying in Germany are often encouraged to return to their countries of origin so that they apply the skills and knowledge there. Similar return-policies do not exist in Canada. In addition, Canadian universities have a relatively flat and transparent faculty career structure that typically progresses from assistant to associate and finally full professor, while German universities are organized much more hierarchically, with research-staff positions (*wissenschaftliche Mitarbeiter*) often being tied to relatively few professorships. Competition for

these professorships is tight and national social capital can play an important role in obtaining one of these highly-desired positions (Behrens et al. 2010). In this context, international mobility may have negative effects if it is associated with a loss of national social capital. Conversely, in Canada, I expect national social capital to be less important due to the Canadian academic field's close integration with the USA and strong links to Britain and the Pacific Rim. Finally, the Canadian portion of the study took place in a part of Canada and at institutions where English, the scientific *lingua franca*, is the working language. Although German universities and research groups increasingly accommodate English, German as the language of everyday interactions (and teaching) may deter internationally mobile academics from locating there.

The sample was collected from a small number of universities and a Max-Planck-Institute located in major metropolitan areas that serve as international knowledge hubs in both countries. To obtain a range of perspectives, the sample of interview participants was stratified by origin country: the Canadian sample includes academics who were educated either in India, other English-speaking countries of the global north, or Germany. The German sample consists of academics educated in India, English-speaking North America, or another European country. The sample was also stratified by gender and career-stage, ranging from doctoral researcher to full professor, with the criteria that a stay abroad has extended beyond one year. In addition, the sample was balanced to equally represent "natural science" disciplines, such as Astrophysics, Cosmology, Biochemistry, or Geology, that often make universal knowledge claims and "social-science" disciplines, such as Anthropology, Business Management, Human Geography, or Political Science, in which knowledge is often contextualized. Research suggests that mobility characteristics differ between disciplines with varying knowledge claims (Crawford et al. 1993; Jöns 2006; Morano-Foadi 2005; Rostan and Ceravolo 2015). Classics, linguistics, and religious studies were allocated to the social sciences; Engineering or Medical disciplines were not represented in the sample. The final sample contains 42 participants, 21 in each country. In the German sample there were 10 men and 11 women, 11 participants in the social sciences and 10 in the natural sciences, and 6 researchers with origins in India, 7 in North America, and 8 in other European countries. In the Canadian sample there were 14 men and 7 women, 11 participants in the social sciences and 10 in the natural sciences, and 8 researchers with origins in India, 6 in Germany, and 7 in other English-speaking countries.

The semi-structured personal interviews were audio-recorded and transcribed. After a pilot involving a small number of interviews and their preliminary assessment, the remaining interviews were conducted. The bulk of the analysis occurred after the interviews were concluded and involved open and axial coding of the transcribed data, carefully assessing the resulting themes and categories for their validity, and interpreting the themes and categories in light of Bourdieu's framework of capital.

In the below presentation of the results, I translated all quotes taken from interviews conducted in German into English. Occasionally, I provide the German term to avoid ambiguity and offer transparency regarding my translation. For all participants quoted, I identify their origin, whether they work in a natural- or social-science

discipline, and their career stage; gender is apparent from the narrative. To preserve anonymity, I cannot reveal the participants' university or research institution, the city in which they are located, or their discipline (other than natural or social sciences).

Results

In this section, I present the results of the analysis in relation to the various narratives of optimistic and pessimistic interpretations developed above. The results based on interviews with participants in both Canada and Germany show that social capital often facilitates international mobility and that international mobility sometimes establishes career-enhancing social capital. Participants affirmed the importance of the national academic field in Germany and noted the loss of national social capital with international mobility. The analysis did not indicate a similar importance of the national academic field in Canada. The analysis further revealed that the connections between international mobility and social capital are complex, multidimensional, and shaped by individual biographies.

Social Capital Facilitates International Mobility

Seven of the Canadian-based participants emphasized that their professional networks and social connections brought them to Canada. For example, a postdoctoral researcher in the natural sciences who received his doctorate in England described how his current postdoctoral supervisor "would come to visit [my previous university] quite frequently actually, and [my doctoral supervisor] would sort of encourage me to go out to dinner with [the visitor] and things like that when there were dinners and if he was speaking and things like that." When a postdoctoral opportunity came up, the doctoral supervisor "very much encouraged me to take [it], I think, his reference letters, which probably helped … And I could get the, sort of, real, inside sort of information … not just like what the department is like, but what [the city in Canada] is like to live in as well." A postdoctoral fellow working in the social sciences who migrated from England to Canada had a similar experience. He explained that his current postdoctoral supervisor "did pop up to see us … he came up to see us for like a workshop, ya, and then we had some email contact about some various things and then um, how did I select him? … Yes, so my supervisor knew him pretty well. Like [my doctoral supervisor] co-edits a journal with him." Similarly, a postdoctoral fellow from Germany now working in Canada in a natural-science discipline explained the importance of networking opportunities with international scholars at his previous institution for obtaining his current position: "so the director of [my current institute] was visiting [my previous institution] and we had a chance to talk there about what I have been doing … [As a result] I got only one offer … from North America which was the position here." For these researchers, social capital in the form of access to a social network was a decisive factor for obtaining a postdoctoral fellowship in Canada and thus in facilitating their international mobility.

In the German sample, 11 participants confirmed that social capital plays a similar role. An early-career researcher who moved together with her partner from postdoctoral positions in North America to work in a natural-science discipline in Germany had originally dismissed the advertisement for her current position "because it was in Germany." However, as the supervisor of both her and her partner declined a job offer in Germany, "he recommended us to the institute here as a kind of way of saying: 'I'm sorry but you know these guys are really good so you know maybe they could fill my shoes to some extent', so we had his support which made a difference here and I think that really helped us get the job at all." In the end, social capital enabled the couple to get jobs at the same institute. An early-career scholar who relocated from North America to Germany to assume a natural-science position also stressed the importance of networking. He explained: "my graduate advisor is a friend with the guy who is the head of this department … So I know him, of course, through my advisor and from being down there." Similarly, a male early-career researcher originally from India remarked that:

> my supervisor in India during [my] Master's, he was already here during his Humboldt fellowship, so he told me, if you go there you will learn something better and he said he will help me also to find a professor. … so he knows some professors … so my professor helped me to search a professor who works in the similar field, so it was the first step.

The importance of social capital in facilitating international mobility – especially among early-career researchers – could be observed among women and men in both Germany and Canada. The networks of supervisors are especially important in this respect, affirming other research (Schiller and Diez 2012). Interestingly, the relationships between researchers and their supervisors could be considered strong ties, while weaker international ties are also involved in facilitating mobility. Unlike my expectation based on the literature (Kyvik et al. 1999), I did not find evidence that the role of supervisors differs in this respect between social- and physical-science disciplines.

International Mobility Develops Social Capital

In the case of a few participants – two in the Canadian and two in German sample – international mobility fostered the accumulation of social capital. In the Canadian sample, an associate professor in the social sciences who received her academic training in India remarked that her previous international experience enabled her to develop networks in India and China. However, she also juxtaposes the value of international networks with that of a domestic network:

> I am also building a network inside Canada because you need to network to survive here. But I think that it has been my international networks that I have used a lot [more] than my Canadian networks but that may change. The way I am thinking of my research moving forward … is also relying very heavily on my global networks.

Interestingly, this participant comments on the importance of a network within Canada to "survive" in the national academic system.

In the German sample, an early-career scientist with a PhD from North America remarked the positive effect of her international mobility that enabled her to network internationally. However, this respondent, too, emphasizes the importance of her domestic network in the context of international mobility: "it's always good to know more people, and so long as the people that I know in the US don't forget about me … then it's always good for a career to know people [in] other places." This participant, too, links the accumulation of international social capital to the value of national social capital in her country of origin.

Both participants establish a link between international mobility, the development of international networks, and the importance of national social capital. They do not see the relationship between international mobility and social capital as unproblematic. I will address the national social capital in greater detail below.

Social Capital Accumulated Abroad Facilitates Career Advancement

The third narrative of the optimistic interpretation of the connection between international mobility and social capital suggests that international mobility fosters career advancement through the accumulation of social capital. This narrative was supported by three participants in the Canadian sample and two in the German sample. A Canadian-based mid-career researcher in a natural-science discipline remarks how the social capital he developed at a stay abroad enabled him to secure a tenured position years later:

> I kept in touch with people at [my current university] over the entire period, seven and a half years and came back once a year … drop in and say hi and then I think they got a large grant … to set up some of this infrastructure that had already existed in the research, scientific research environment, the business environment and they were looking for someone to manage that. Basically that's where they came knocking on my door and asked me if I would be willing to start as sort of a junior professor here, assistant in a non-tenure track position.

After three years in this position, the participant was offered a tenure-track position. At the point of the interview, he has been a tenured professor for five years at this institution.

An example from the German sample is that of an early-career scholar in a natural-science discipline who obtained his training in another European country. He received an invitation for a previous postdoc in the USA "through contacts" and that this position, in turn, enabled him to develop many more contacts that increased his scholarly productivity. He said:

> I still have this contact to the people that I got to know then. I still work with them. I'm now doing lots of projects with them. So I didn't do anything with them for a long time but then someone contacts me and then we do a project together. Mobility was hugely important for such contacts.

Although this participant did not directly speak of career advancement, enhanced research productivity is typically recognized in hiring and promotion processes, and thus related to career advancement. In this way, the accumulation of social capital through international mobility leads to the accumulation of other forms of capital signifying value in the academic labor market. Although both the Canadian and German samples provided evidence in support of the narrative that social capital accumulated abroad facilitates career advancement, only a few participants had such experiences.

National Academic Field

Many participants emphasized the importance of the national academic field. 11 participants in the German sample commented on the nature of the national academic field in the context of social capital. A mid-career scholar from another European country who works in a social-science discipline is participating in a program helping female scholars from abroad to establish scholarly networks to facilitate their integration into the German academic system. She remarked:

> You notice that these programs are necessary because in Germany it is much more the case that you don't advance without contacts. Unfortunately, that's often the case. In [my country of origin] it plays a role too, of course, everywhere it plays a role, but I don't have the sense that it is shaped (*geprägt*) to such a degree by personal contacts.

A professor in a social-science discipline who received her academic credentials in another European country explained: "When you are here [in Germany], you establish a network, you have a scholarly (*wissenschaftliches*) network, then you stay here because you know all the colleagues." Overall, the interviews revealed the importance of national social capital in the German academic field.

Respondents in the German sample generally recognize the importance of national social capital for career advancement within Germany. An early-career researcher in the social sciences who received his academic qualifications in North America remarked:

> I had a couple of [job] offers [in Germany] … they were both from people I knew who I'd met at conferences, who I had interacted with at some professional level or these people at least knew my supporters here I mean I had a very, a very kind mentor who I think did some work for me and that helped immensely; you know, I don't think it's exaggerated to say that these things work at a very personal level, at least in my field in Germany.

When asked about the career advantages of her current position, a postdoctoral researcher in a natural-science discipline who came to Germany from North America replied: "it's always important like not only who you know, but [who] your advisor and the people you work with know because they are going to be hopefully recommending you for other positions." The relationships to an advisor and colleagues resemble strong ties; when asked about his career progression, an early-career

researcher in the natural sciences who came from another European country explained the importance of weak ties:

> It won't be easy …. My discipline is very small, there are not many positions; you can't be too picky. The question is, do you know people there [at another institution in Germany], is there a group which you can join. … If there is a job advertised, the first question is if I know someone there and if there is a possibility. Usually the first question is, do you know someone there. That is the strategy. … It helps to know people.

In the Canadian sample, the Canadian national academic field was not a prominent theme in the interview-discussions related to social capital. When there were comments, they referred to other national academic fields. For example, a postdoctoral fellow from India who works in the natural sciences intends to mobilize the social capital she accumulated in India – consisting of connections to former professors, fellow students, and "their networks" – to return to India.

Loss of National Social Capital

The pessimistic interpretation of the relationship between international mobility and social capital focuses in particular on the loss of national social capital while abroad. Four of the six German-origin academic researchers in the Canadian sample commented on this issue. A Canadian-based postdoc in a natural-science discipline explained the impact of being away from Germany:

> [My former university in Germany] just recently hired somebody who is actually very related to what I am doing so in reality there would have been maybe a job opportunity there which I didn't know about – I know people there, and they know me so in some sense it shows that I was very strongly out of the system there, meaning that they didn't think about me and said "oh why don't you apply for this" or so. So I'm starting to see that, in terms of my German roots and being in Germany, I'm cutting myself a bit. So I think in that sense I have actually slightly harmed myself. … I realize that I am falling out of the system, meaning that I'm not anymore [somebody] you know.

A German-origin professor in the social sciences reflected in a similar way on his experience of being away from the German academic field for extended periods:

> In retrospect, I think that was the great disadvantage of my frequent movement because in the first six or seven years, because I always thought that I will go back, I did not really establish a network here. By the time I left Germany and decided I will stay here permanently, I more or less made a sharp cut with Germany and my German past, so I lost my network in there.

Three participants in the German sample comment in similar ways about their German colleagues who decided to go abroad. For example, a professor in a social-science discipline with North-American scholarly roots said:

> My impression in Germany, especially in the small field of studies, is that networking is the most important thing, and it's very, very hard for academics outside to break into this. I even know German scholars of [his field] who have simply been away way too long and have trouble getting back in, you know; it really depends on who you know and if they know you are looking for a job.

These quotes illustrate that in the German academic field, being abroad is often associated with the loss of national social capital – especially weak ties – and, in this way, erects a barrier to return to Germany.

The loss of national social capital through international mobility is not restricted to the German academic field. One assistant professor in a social-science discipline with Indian roots who worked in Britain and India before moving to Canada remarked:

> It is hard to maintain the networks [in Britain and India] and, in some ways, one ends up dropping them. … My strongest networks for a while really had been in Britain and in India. They continue to be very strong in India, but because I can spend less time there now, you know having a family here, I think, now I am developing stronger networks to supplement those in the United States and in Canada.

In the same vein, one Canadian-origin postdoc in a natural-science discipline who now works in Germany explained that, on the one hand, networking is helping him accumulate the social capital required in Germany for his career advancement; on the other hand, his absence from Canada diminishes his opportunities there:

> The point is that I am having problems in finding the job from a post-doc to a group leader and I don't know if the networking, the international networking, is helping in this step because I think what is important for me at this time is that I have to get hired, to be hired here in [Germany] and the networking inside the university here is what is important … . And I think it's the same if I am going back to Canada: I think I lost a lot of my contacts inside the Canadian universities and so if opportunities come up I don't think I am on people's minds.

Interestingly this participant notes the loss of national social capital in Canada as a result of international mobility although the preceding section on the academic national system implied that national social capital is not a major issue in the Canadian context.

The relevance of social capital also extends beyond the academic field. One Canadian-based Indian-origin professor in the natural sciences emphasized his local network with national industry partners:

> Here it's important to have contacts in industry because we work very closely with industry. Trying to get research funding from industry is a very big part of our job as well. So if I were to move to the US, for example, losing those

> types of contacts would have hurt me more than just academic networks. … If [I] would move back to India that is something I would have to start over again because I don't have those sort of connections and it's not easy building that overnight because those are the sort of ties that develop working together, they have to become familiar with you, they have to see some of your work, you have to work with them, so it actually takes years.

Networks extending to national industry partners adds another layer to the place-bound nature of social capital and the possible loss of social capital with international mobility.

There were also comments from participants that challenge the narrative that international mobility is associated with a loss of national social capital. One Canadian-based German-origin mid-career scholar in a social-science discipline explains that his network in Germany actually improved since moving to Canada. When asked if he had difficulties maintaining his network in Germany, he said:

> No, not at all, rather the contrary … I still am in touch with people over there, … I'm also teaching at the Centre for [field of study], and one of the things I initiated with a colleague of mine the past years, was kind of a student, graduate student exchange, or kind of a graduate student workshops with [two German universities]. We are actually using my contacts, and I use them now, for example, in this case to bring students and colleagues to Germany and Germans to come over here. … I don't feel that I lost the network, rather the opposite.

One Canadian-based German-origin postdoctoral researcher in a natural-science discipline challenges the idea that developing social capital within the national academic field in Germany improves career prospects. He explained: "So I fell out of the German system in particular which is where I sort of was growing up until I basically came here. All my scientific career was in Germany and that has not lasted or helped me to be more visible in Germany." While it is important to acknowledge these counter positions, more participants suggested that international mobility is associated with the loss of national social capital.

Discussion

The findings of this paper correspond with the literature suggesting that the connections between the international mobility of academic researchers and social capital are multidimensional. International mobility can have positive and negative effects on the accumulation of social capital and subsequently other forms of capital that contribute to career advancement. Among participants in both the Canadian and German samples, social capital often facilitated international mobility. Early-career academics especially benefited from the strong ties to their supervisors, and the strong and weak professional ties their supervisors had cultivated internationally. In some cases, international mobility, in turn, also enhanced the accumulation of social capital. Interestingly, the responses of several

participants suggest that social capital was not accumulated strategically for the purpose of international mobility. Rather, international mobility was an unintentional result of existing social capital, for example, when supervisors encouraged early-career academics to assume positions abroad. This finding complements research showing that career development, especially in the early-career stages, is part of a deliberate career building strategy (Jung 2014; Khattab and Fenton 2016; Lee and Kuzhabekova 2018).

The findings show that social capital accumulated abroad can lead to increased productivity and career advancement. Although in the sample of this particular study only few participants had such experiences, this finding corroborates research on mobile academics in Hong Kong and Macau showing that transnational job mobility has positive effects on research output in STEM fields (e.g., Horta et al. 2017). Recent research of Italian academics, however, also suggests that early-career scholars who move internationally to prestigious institutions were already high performers before they moved (Cattaneo et al. 2018). Although the design of the current study did not capture the degree to which participants could be considered elite, high, average, or below-average performing scholars, the evidence suggests that, in some cases, international mobility is associated with positive career outcomes related to social capital. The findings also affirm the importance of structural inequalities of the international academic system (Jöns 2015; Bauder et al. 2018), with participants who have gained experience in the academic core (e.g., the USA) espousing optimistic views of international mobility, while scholars with ties extending into peripheral countries such as India and China focus on building networks within Canada.

Many participants also emphasized negative effects of international mobility: the value of social capital to secure employment is often restricted to the national academic field. As a result, international mobility frequently isolates academic researchers from national networks and thus domestic career opportunities in the origin country. Many interview participants stressed the importance of national social capital especially in the German context and suggested that hiring practices at German universities valorize national social capital relative to scholarly merit and competencies. While these practices may be historically entrenched in the German academic field, they have recently been intensely criticized (Behrens et al. 2010). Interestingly, even international experience in a place like the USA, which represents the core of a hierarchical international academic system (Bauder et al. 2018), may not be an advantage relative to the possible loss of national social capital. Unless the situation of a relatively closed German labor market and the need to maintain national social capital is rectified, the internationalization strategies of many institutions and the valorization of "internationality" (Hamann and Zimmer 2017), especially among many early-career researchers (Röbken 2009), seems counter-productive. This situation raises questions for policymakers and managers at institutions, such as Germany's Alexander von Humboldt Foundation, to encourage international mobility among early-career researchers. At the same time, however, it explains the need to establish programs and networks, such as the German Academic International Network (GAIN), helping internationally-mobile early- and mid-career scholars to return after longer periods abroad.

Although national social capital will likely remain relevant in faculty hiring and career advancement in some countries (Behrens et al. 2010; Combes et al. 2008), with the increasing internationalization of university systems, international social capital may complement or even take the place of national social capital in providing access to jobs and prestige. This prospect raises questions about the role of the "invisible college" in professional reproduction: the analysis found that international social capital – often in the form of strong ties among supervisors and within departments – facilitates international mobility; the analysis also provided some evidence that international mobility creates social capital and enhances career advancement. More research is needed to understand if and how international "invisible colleges," consisting of international research groups and communities, reproduce themselves when early-career researchers accumulate social capital through their supervisors and departments that enables mobility, which, in turn, provides access to career opportunities. Whether the academic field is national or international in scale, the lack of social capital may exclude competent and meritorious academic researchers from prestigious jobs and research projects. In particular, early career researchers may be affected by this exclusion based on the lack of social capital. Conversely, *who* early-career researchers know in an international network may be critical for career advancement.

This study had several important limitations that could be addressed in future research. First, while the study emphasized the national character of the academic field in which social capital has value, it did not explore the importance of language communities in this respect. For example, many German-origin academic researchers work in German-language universities in Switzerland (Van Noorden 2012), which raises questions about the linkages between international mobility and social capital within language communities. Mobility between English-speaking Canada and the USA vs. French-speaking Quebec may provide equally interesting case studies.

Second, although this study collected information on gender and origin, the data did not provide any conclusive evidence of corresponding differences in respect to the experiences of how international mobility and social capital are connected, as one might have expected based on the literature (Bagilhole and Goode 2001; Davis 2001; Röbken 2009; Lutter and Schröder 2016). This outcome may relate to this paper's relatively narrow focus on employment-related mobility and social capital among academic researchers. Future research could take a wider lens and consider family relations, work-life balance, and the benefits of mobility to family and non-professional life.

Third, the sample does not include cases of return mobility. At the time when the study was designed or when the pilot interviews were completed, I did not anticipate the negative impact of national social capital. In retrospect, it would have been interesting to include return migrants to explore how they deal with the loss of national social capital or how they may have been able to maintain national social capital. Future research with a different sample design could address this limitation.

Finally, this study's sample of 42 is relatively small and not representative of the academic community as a whole. In addition, only a few participants experienced that international mobility creates productive social capital or enhances careers.

Small numbers *per se* do not necessarily diminish the validity of the findings. For example, in a different study only about 10% to 20% of Swedish researchers who went abroad for a postdoc had negative experiences upon their return (Melin 2005); the small percentage does not diminish the findings – for this relatively small group these negative experiences are undoubtedly important – but says something about the magnitude of this particular problem. A possible limitation may relate to the sequence of data collection and analysis: some nuances in the findings presented above were uncovered only in the context of the in-depth analysis and were not apparent in the pilot phase or while the interviews were conducted. Future research with a larger sample and/or quantitative research design should assess the magnitude of the various aspects unveiled in this study that connect international mobility and social capital.

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